

SMART INDIA HACKATHON • SOFTWARE EDITION

AksharaPlay: Low-Cost Dyslexia Screening & Multi-Lingual Game Suite

Early primary school diagnostic ecosystem powered by edge speech-recognition and NLP

Theme: Smart Education / Inclusive Tech

Category: Software

Team ID: [Insert Team ID]

Idea Title & Proposed Solution

Proposed Solution

A gamified mobile application disguised as an immersive phonics adventure game that screens children for dyslexia during early primary school years.

Core Value: Bypasses clinical psychologist shortages by shifting preliminary diagnostic tracking directly into standard school classrooms.

3-Tiered Architecture

- 1. Gamified Tracking:** Captures touch latency and erratic visual saccades during gameplay.
- 2. Speech & NLP:** Analyzes phoneme substitution and syllable omission in regional Indian languages.
- 3. Teacher Dashboard:** Converts metrics instantly into actionable clinical risk reports.

Technical Approach



Frontend & Game

Flutter cross-platform mobile & tablet framework built for public school device compatibility.



Speech Engine

Offline TensorFlow Lite & Vosk API running locally to evaluate regional phonemes without cloud latency.



Backend & Analytics

Python FastAPI paired with scikit-learn classification models and secure PostgreSQL caching.

Feasibility and Viability

Technical Feasibility

Built using mature open-source audio frameworks, allowing rapid prototyping and local offline execution in remote rural classrooms.

Risk Mitigation

Noise Filtering: Integrated noise suppression filters and headset prompts during voice checks.

Child Fatigue: Wrapped clinical metrics inside rewarding fantasy levels to eliminate test anxiety.

Impact and Benefits



Social Inclusion

Stigma-free early screening for rural children, preventing kids from being mislabeled as slow learners.



Cost Efficiency

Eliminates heavy financial barriers of private clinical assessments for low-income families.



National Alignment

Directly fulfills foundational literacy and inclusive education mandates under the NEP framework.

Research and References

#	Reference Framework / Source	Core Contribution to Project
1	National Education Policy (NEP)	Foundational literacy, numeracy, and inclusive school ecosystem guidelines.
2	Dyslexia Association of India (DAI)	Statistical data on uncertified learning disabilities and bottlenecks.
3	Multilingual Speech Standards	Acoustic modeling literature for low-resource regional Indian phonemes.
4	Gamified Cognitive Case Studies	Frameworks for non-invasive neuro-developmental mobile tracking.